

Flexural Behaviour of Uncracked PC Members - Additional Examples and Exercises

Example 1

A simply-supported beam with a cross section as shown in Fig 12 spans 15 m and carries a total uniform load of 30 kN/m (excluding self-weight). If the beam is prestressed with a force of 2,000 kN acting at an eccentricity of 400 mm below the centroid,

- Determine the stress distribution at mid-span during both transfer and service stages. You may assume that the loss at transfer and service are 15% and 25% respectively.
- Calculate the line of pressure at the mid-span at service.
- Draw the line of pressure for the whole beam at service.

It is given that the cross section area of concrete is 290,000 mm² and the elastic section moduli at top and bottom fibres are the same equal to 70.73×10⁶ mm³. Density of concrete (with prestressing steel) = 24 kN/m³.

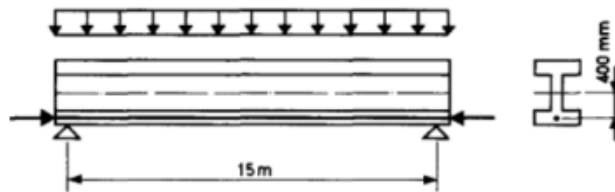


Fig 12 A simply-supported beam carrying a uniformly distributed load

Solution:

(i)

$$\text{Self-weight of beam} = 24 \times 290000 \times 10^{-6} = 7 \text{ kN/m}$$

Maximum bending moment at mid-span

At transfer:

$$\frac{wL^2}{8} = \frac{7 \times 15^2}{8} = 197 \text{ kNm}$$

At service

$$\frac{wL^2}{8} = \frac{(30 + 7) \times 15^2}{8} = 1041 \text{ kNm}$$

Effective prestressing forces

$$\text{At transfer} = 2000 \times 0.85 = 1700 \text{ kN}$$

$$\text{At service} = 2000 \times 0.75 = 1500 \text{ kN}$$

Stress distribution

At transfer, top surface:

$$\sigma_T = \sigma_D + \sigma_{Pe,t} + \sigma_{B,t}$$

$$= \frac{P}{bh} - \frac{Pe}{Z_t} + \frac{M_{SW}}{Z_t}$$

$$= \frac{1700 \times 10^3}{290000} - \frac{1700 \times 10^3 \times 400}{70.73 \times 10^6} + \frac{197 \times 10^6}{70.73 \times 10^6} = -0.96 \text{ MPa}$$

At transfer, bottom surface:

$$\sigma_B = \sigma_D + \sigma_{Pe,b} + \sigma_{B,b}$$

$$= \frac{P}{bh} + \frac{Pe}{Z_b} - \frac{M_{SW}}{Z_b}$$

$$= \frac{1700 \times 10^3}{290000} + \frac{1700 \times 10^3 \times 400}{70.73 \times 10^6} - \frac{197 \times 10^6}{70.73 \times 10^6} = 12.7 \text{ MPa}$$

At service, top surface:

$$\sigma_T = \sigma_D + \sigma_{Pe,t} + \sigma_{B,t}$$

$$= \frac{P}{bh} - \frac{Pe}{Z_t} + \frac{M_t}{Z_t}$$

$$= \frac{1500 \times 10^3}{290000} - \frac{1500 \times 10^3 \times 400}{70.73 \times 10^6} + \frac{1041 \times 10^6}{70.73 \times 10^6} = 11.4 \text{ MPa}$$

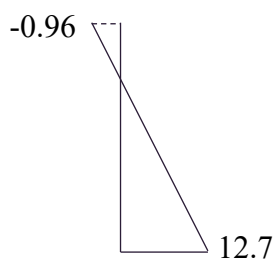
At service, bottom surface:

$$\sigma_B = \sigma_D + \sigma_{Pe,b} + \sigma_{B,b}$$

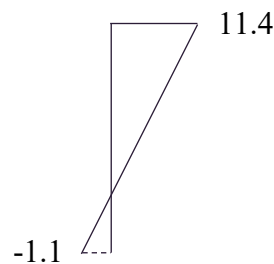
$$= \frac{P}{bh} + \frac{Pe}{Z_b} - \frac{M_t}{Z_b}$$

$$= \frac{1500 \times 10^3}{290000} + \frac{1500 \times 10^3 \times 400}{70.73 \times 10^6} - \frac{1041 \times 10^6}{70.73 \times 10^6} = -1.1 \text{ MPa}$$

Stress distribution diagram for the cross section



At transfer



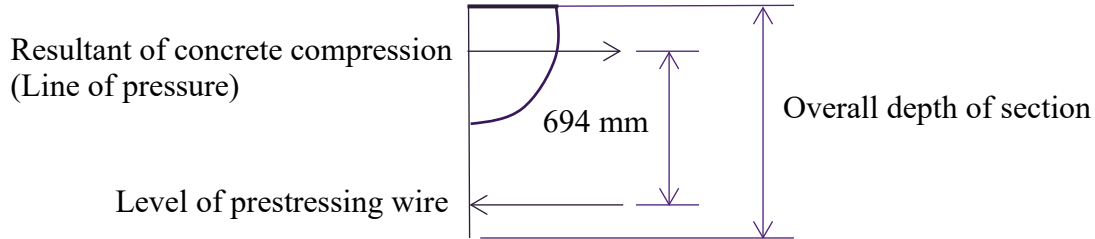
At service

(ii)

During service at mid span, the moment is 1,041 kNm and the effective prestressing force is 1,500 kN. By Eq (8), the lever arm z can be evaluated as:

$$z = \frac{M}{P} = \frac{1041 \times 10^6}{1500 \times 10^3} = 694 \text{ mm}$$

Therefore, the resultant compressive force contributed by concrete is located at 694 mm from the prestressing wire.



(iii)

The variation of bending moment creating by the uniformly distributed load w can be described by the following second-order polynomial equation:

$$M = -\frac{wx^2}{2} + \frac{wL}{2}x$$

where w is the intensity of the uniformly distributed load, L is the span length of the beam and x is the distance from the end support. In this example, $w = 37 \text{ kN/m}$ and $L = 15 \text{ m}$, and therefore the bending moment equation is:

$$M = -18.5x^2 + 277.5x$$

The line of pressure profile can be obtained similarly by Eq (8), where

$$z = \frac{M}{P} = \frac{-18.5x^2 + 277.5x}{1500} = -0.01233x^2 + 0.185x \text{ (in m)}$$

From the above, it is evident that since there is externally applied moment, the line of pressure will follow the shape of the external bending moment diagram, which is different from the tendon profile. However, if there is no externally load applied, it is evident from Eq (8) that the lever arm is always zero indicating that the line of pressure is coincident with the tendon profile. Technically, a tendon profile having this property is named a **concordant profile**. The bending moment diagram generated by external load is one of the concordant profiles.

Example 2

A simply-supported beam of length L has a parabolic tendon profile with maximum eccentricity e , as shown in Fig 13. Determine the upward force on the beam exerted by the tendon and draw the shear force and bending moment diagrams due to prestress.

$$y = 4ex(L - x)/L^2$$

where e is the maximum eccentricity at the mid-span, L is the span length and x is the distance from either end support.

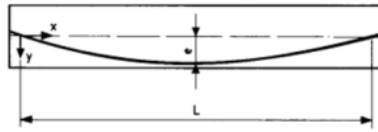


Fig 13 A simply-supported beam with parabolic tendon profile

Solution

Assume $e \ll L$. For small eccentricity compared with the span length, the curvature of the profile can be approximated by $\frac{1}{r} = \frac{d^2y}{dx^2}$, where r is the radius of curvature. Therefore the curvature of the given parabolic profile can be obtained by differentiating the parabolic profile twice with respect to x , which gives:

$$\frac{d^2y}{dx^2} = -\frac{8e}{L^2}$$

From Eq (9b), it is noted that the equivalent load acting on the concrete due to curvature of the profile can be determined by multiplying the prestressing force to the curvature of the profile. Therefore,

$$w = P \left(\frac{d^2y}{dx^2} \right) = P \left(-\frac{8e}{L^2} \right) = -\frac{8Pe}{L^2}$$

Then the shear force and bending moment diagram can be obtained by the principle of elementary beam bending theory:

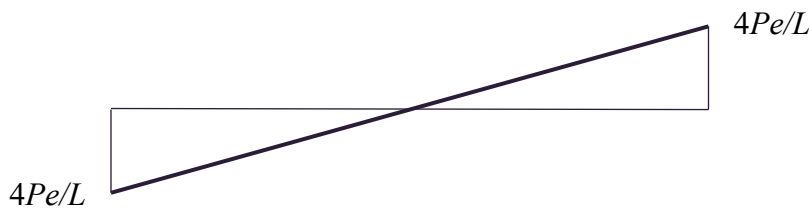
Variation of shear force

$$V = -wx + \frac{wL}{2}$$

Substituting $w = -8Pe/L^2$, we have:

$$V = \frac{8Pex}{L^2} - \frac{4Pe}{L}$$

Maximum shear occurs at both ends of beam, which is equal to $4Pe/L$



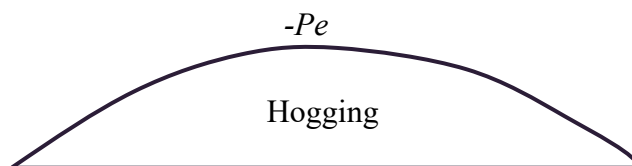
Variation of bending moment

$$M = -\frac{wx^2}{2} + \frac{wL}{2}x$$

Substituting $w = -8Pe/L^2$, we have:

$$M = \frac{4Pex}{L} \left(\frac{x}{L} - 1 \right)$$

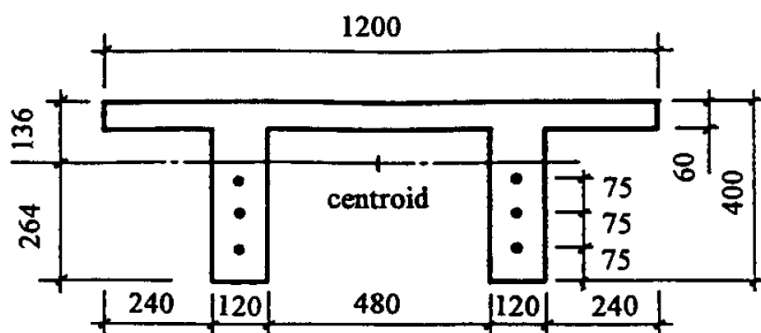
Maximum bending moment at mid-span, which is equal to $-Pe$ (Hogging)



Exercise 1

The precast prestressed concrete double-T beam as shown in Fig 14 is designed for an imposed load of 6 kN/m over a simply-supported span of 10 m. The beam is pre-tensioned with six 12.9 mm diameter strands with constant eccentricities, each being stressed initially to 70% of the characteristic strength. The beam is lifted shortly after transfer using the lifting eyes provided at the quarter points.

- Check the stress conditions during lifting.
- Check the stress conditions under working load.



$$A = 0.1536 \text{ m}^2$$

$$I = 0.002338 \text{ m}^4$$

Fig 14 Cross section of double-T beam
(all dimensions are in mm)

Given the following material properties:



Concrete

Characteristic strength = 50 MPa

Strength at transfer = 40 MPa

Density of concrete (with steel) = 24 kN/m³

Steel

Characteristic load of 12.9 mm diameter strand = 186 kN

Nominal steel area of 12.9 mm diameter super strand = 100 mm²

Loss of prestress

At transfer = 15%

At working load = 25%

[Numerical answers:

(a) At transfer, critical point at quarter points: top = -0.75 MPa; bottom = 14.2 MPa

(b) At working, critical point at mid-span: top = 7.0 MPa; bottom = -2.3 MPa]

Properties of Materials-Examples

Example 3

Referring to Example 1 above, select an appropriate concrete cylinder strength f'_c . You may, for simplicity, assume that $f_{cm} = 1.1(f'_c + 3)$ MPa, $f_{ct,f} = 0.6\sqrt{f'_c}$ and f_{cp}' (at transfer) $= 0.6 f'_c$.

Solution

Referring to the stress limits stated in Section 2.4, the selected value concrete strength should satisfy the following conditions:

At transfer

$$\text{Top fibre: } 0.96 \text{ MPa} \leq 0.5 \times 0.6 \sqrt{f_{cp}'} \Rightarrow f_{cp}' \geq 10.2 \text{ MPa}$$

$$\text{Bottom fibre: } 12.7 \text{ MPa} \leq 0.5 \times 1.1(f_{cp}' + 3) \Rightarrow f_{cp}' \geq 20 \text{ MPa}$$

Therefore at transfer, $f_{cp}' \geq 20 \text{ MPa} \Rightarrow f'_c \geq 20 / 0.6 = 33.3 \text{ MPa}$ (at 28th day)

At service

$$\text{Top fibre: } 11.4 \text{ MPa} \leq 0.5 \times 1.1(f'_c + 3) \Rightarrow f'_c \geq 17.8 \text{ MPa}$$

$$\text{Bottom fibre: } 1.1 \text{ MPa} \leq 0.5 \times 0.6\sqrt{f'_c} \Rightarrow f'_c \geq 13.4 \text{ MPa}$$

It is evident that the design of concrete strength is governed by the maximum tensile stress at top fibre at transfer. The selected f'_c should be $\geq 33.3 \text{ MPa}$ (say 40 MPa – the closest standard strength grade).

Example 4

Determine the shrinkage and creep coefficient of a concrete specimen of $f'_c = 40 \text{ MPa}$ and with a hypothetical thickness of 200 mm after 10 years. For creep strain evaluation, you may assume the sustained stress is 15 MPa.

Shrinkage

Final autogenous shrinkage:

$$\varepsilon_{cse}^* = 50 \times 10^{-6} (0.07 \times 40 - 0.5) = 115 \times 10^{-6}$$

Autogenous shrinkage strain:

$$\varepsilon_{cse} = \varepsilon_{cse}^* (1.0 - e^{-0.07t}) = 115 \times 10^{-6} (1 - e^{-0.07 \times 365 \times 10}) = 115 \times 10^{-6}$$

Basic drying shrinkage:

$$\varepsilon_{csd.b} = (0.9 - 0.005 \times 40) \times 800 \times 10^{-6} = 0.00056$$

Drying shrinkage strain:

$$\varepsilon_{csd} = k_1 k_4 \varepsilon_{csd.b} = 1.19 \times 0.6 \times 0.00056 = 400 \times 10^{-6}$$

(Note $k_1 \approx 1.19$ as shown in Fig 1)

Total shrinkage after 10 years:

$$\varepsilon_{cs} = \varepsilon_{cse} + \varepsilon_{csd} = (400 + 115) = 515 \times 10^{-6} \quad (3)$$

Creep

Basic creep coefficient $\varphi_{cc.b} = 2.8$ for $f'_c = 40$ MPa

Design creep coefficient:

$$k_3 = \frac{2.7}{1+\log(\tau)} = \frac{2.7}{1+\log(28)} = 1.103 \quad k_4 = 0.6$$

$$k_5 = 1.0$$

$$k_6 = 1.0$$

$$k_2 = 1.18 \text{ (from Fig 2)}$$

$$\varphi_{cc} = k_2 k_3 k_4 k_5 k_6 \varphi_{cc.b} = 1.18 \times 1.103 \times 0.6 \times 1.0 \times 1.0 \times 2.8 = 2.187$$

Creep strain after 10 years:

$$\varepsilon_{cc} = \frac{\varphi_{cc} \sigma_0}{E_c} = \frac{2.187 \times 15}{32800} = 1000 \times 10^{-6}$$

Problem 3 – Calculation of stresses using equivalent load method

The cross-section of a post-tensioned concrete beam spanning 15 m, simply supported, is shown in Figure 3.1. Beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 300 mm. The tendon comprises of 8 number 12.7 mm strands. Tendon is stressed to 80% of breaking load. Properties of the materials are:

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

$$\text{specific weight of concrete} = 25 \text{ kN} / \text{m}^3$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$f_{py} = 1640 \text{ MPa}$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

Using the equivalent load method determine:

1. Stresses in the beam section at midspan due to prestress acting alone
2. Stresses at the midspan section due to prestress, self-weight and uniform live load of 18 kN/m.
3. Is the beam cracked under this loading?

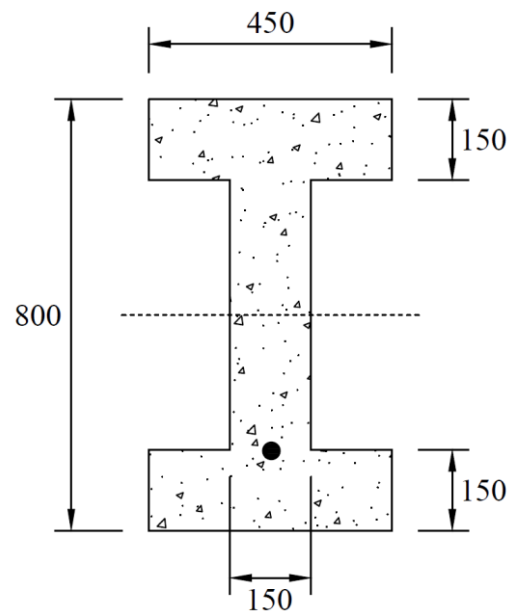


Figure 1.1 - Post-tensioned beam section



Problem 4 – Stresses in prestressed one-way slab using equivalent load method

A 220 mm thick slab spans 10 m. It will be post-tensioned with regularly spaced tendons with parabolic profiles and zero eccentricity at end anchorages. To limit the deflection of the slab, the designer has decided to balance the self-weight of the slab completely by the applied prestress.

Material properties are:

$$f'_c = 50 \text{ MPa}$$

$$f'_{cp} = 40 \text{ MPa}$$

Strands of nominal diameter 12.7 mm

$$A_p = 800 \text{ mm}^2$$

$$f_p = 1870 \text{ MPa}$$

$$f_{py} = 1640 \text{ MPa}$$

Tendon stressed to 70% of breaking load.

Assuming that the minimum cover required for the tendon is 25 mm, and that immediate losses are 5% of the jacking force and time-dependent losses are 17% of the force at transfer;

1. Find the prestress force required to balance the self-weight
2. Check whether the slab can sustain the applied prestress at transfer.